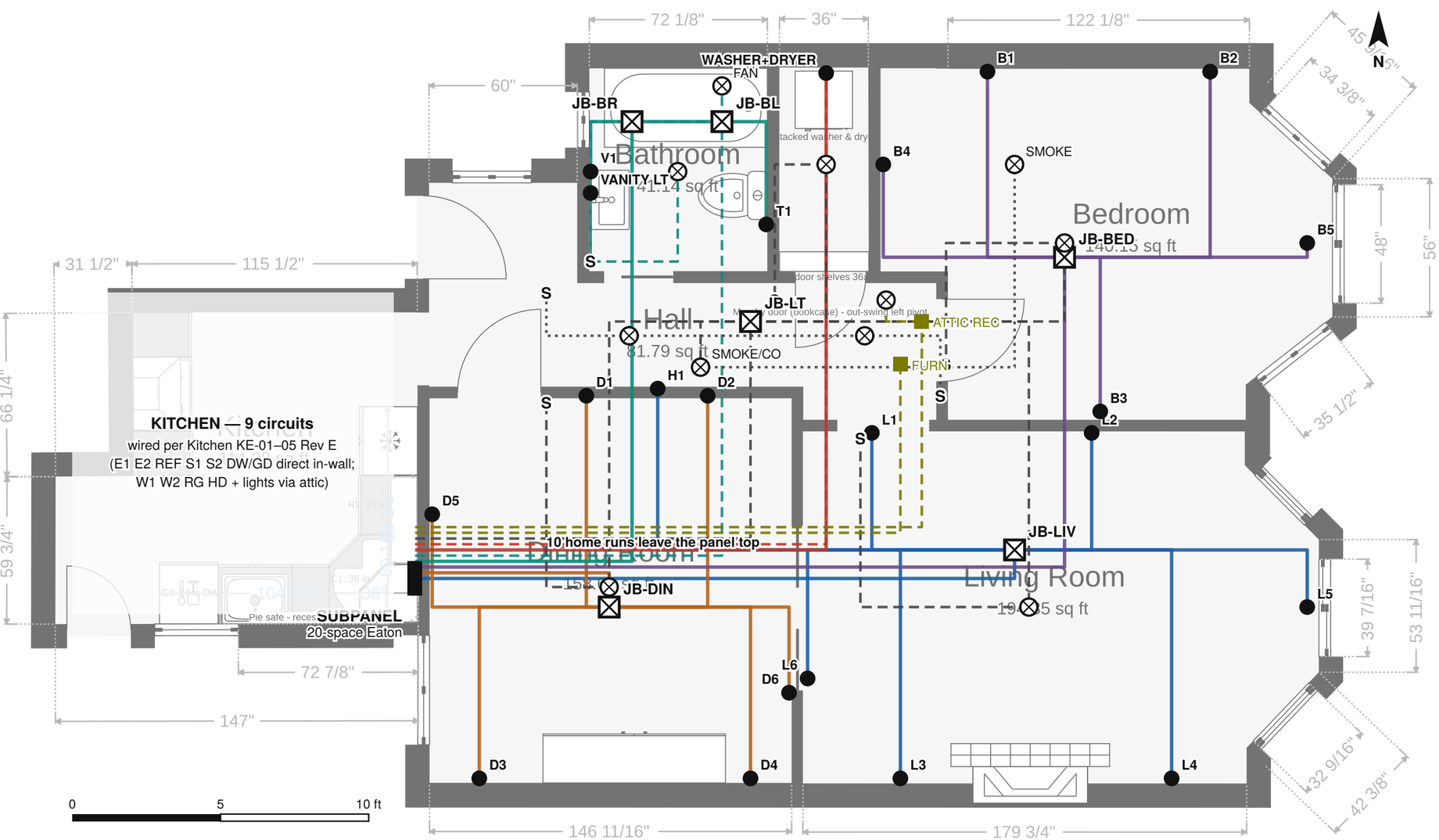
Unit 4 — Whole-unit branch wiring from the subpanel (attic routing)

1231 11th Ave, Unit 4 · Sheet E-W1 · Rev R1 · 2 Nov 2026 · WORKING PLAN for our crew — not a permit set · Scale 1/4" = 1'-0" on 11×17



HOW IT RUNS

Every circuit leaves the top of the subpanel, goes up the stud bay into the attic, then runs to one junction box (JB) per room group. From each JB, one cable drops down the wall to each outlet. Lines on this sheet are the ATTIC runs; the drop is straight down.

CIRCUIT COLORS (labels govern in B/W)

- LIV
- DIN
- BED
- BATH-R / BATH-L
- LAUN-W / LAUN-D
- LTG lights + alarms
- HVAC / ATTIC

CABLE

- 12/2 — solid line (20 A circuits)
- 14/2 — dashed (15 A circuits)
- 14/3 — dotted (hall 3-way, alarm link)

SYMBOLS

- Outlet (wall box)
- S Switch
- Ceiling box (light / fan / alarm)
- Attic junction box, covered + accessible
- Subpanel (D04, faces dining)

NOT ON THIS SHEET

Exact box marks: use the room drawings (Walls/Trim/Outlets R1, bedroom packet, kitchen KE Rev E). Positions here are placed to the right wall, not field-marked. Furnace + hatch location: placed in hall attic as an estimate — move to actual.

Base plan: Magicplan Unit 4 export (26 Oct). Kitchen fixtures shown under the white box are the old layout — ignore them.

ROMEX TO BUY — whole unit (Unit 4 only)

12/2 (yellow) — 20 A	679 ft → 815 ft	**4 × 250 ft**
14/2 (white) — 15 A	558 ft → 670 ft	**3 × 250 ft**
14/3 — hall 3-way, alarm link	95 ft → 115 ft	**2 × 100 ft**

Plan lengths incl. 18 in tails per box + 20% extra (joist direction). Buy 250-ft rolls; return unopened rolls. Kitchen from KE Rev E. Copper NM-B with ground. Separate 100 A feeder is not included.

ATTIC JUNCTION BOX RULES

- Yes: one circuit can split to many outlets at an attic JB.
- Box stays accessible — cover never buried by insulation or flooring; mark it with a flag or tall stake (314.29).
- Screw the box to a joist/rafter; blank cover on (314.23, 314.28).
- Use 4-11/16" sq × 2-1/8" metal boxes (42 cu in); JB-LIV (8 cables) add a 1-1/2" extension ring. One circuit per JB.
- Staple cable within 12" of every box, every 4-1/2 ft (334.30).
- Within 6 ft of the hatch, cable across joist tops needs guard strips — or run along the joist sides (334.23→320.23).
- Max 2 cables per bored hole in the top plate above the panel; seal holes with fire caulk.

Unit 4 — Whole-unit circuit schedule and parts (Sheet E-W2, R1)

1231 11th Ave, Unit 4 · 2 Nov 2026 · Working plan for our crew, goes with routing sheet E-W1. Not a permit set. Subpanel: Eaton BRP20L125G (20 full-size spaces), in the old fridge-wall doorway at D04, facing dining.

Circuit schedule — 19 of 20 spaces used

Space	Circuit	Serves	Breaker	Breaker type	Cable	Route
1	SA-A	East counter E1, E2	20 A	Dual-function	12/2	Kitchen KE Rev E
2	SA-B	South counter S1, S2	20 A	Dual-function	12/2	Kitchen KE Rev E
3	SA-C	West counter W1, W2 (via attic)	20 A	Dual-function	12/2	Kitchen KE Rev E
4	REF	Refrigerator	20 A	Dual-function	12/2	Kitchen KE Rev E
5	DW	Dishwasher (sink-base 2-gang)	15 A	Dual-function	14/2	Kitchen KE Rev E
6	GD	Disposal, switched (sink-base 2-gang)	15 A	Dual-function	14/2	Kitchen KE Rev E
7	RG	Gas range (via attic)	15 A	Dual-function	14/2	Kitchen KE Rev E
8	HD	Hood, hardwired (via attic)	15 A	AFCI	14/2	Kitchen KE Rev E
9	LT	Kitchen main + task lights (via attic)	15 A	AFCI	14/2 + 14/3	Kitchen KE Rev E
10	HVAC	Furnace (Trane S8X1A040M3PTCC) via disconnect	15 A	standard, per Trane	14/2	panel → attic → straight to box
11	LIV	Living outlets L1–L6 + hall H1	20 A	AFCI	12/2	panel → attic → JB-LIV → drops
12	DIN	Dining outlets D1–D6 (small-appliance, no lights)	20 A	AFCI	12/2	panel → attic → JB-DIN → drops
13	BED	Bedroom outlets B1–B5	20 A	AFCI	12/2	panel → attic → JB-BED → drops
14	BATH-R	Bath outlets V1 + T1 (bath only)	20 A	GFCI	12/2	panel → attic → JB-BR → drops
15	BATH-L	Bath ceiling light, vanity light, exhaust fan	15 A	standard (AFCI ok)	14/2	panel → attic → JB-BL → drops
16	LAUN-W	Washer receptacle (laundry 2-gang, left)	20 A	Dual-function	12/2	panel → attic → straight to box
17	LTG	Lights: hall, living, dining, bedroom, laundry + smoke/CO alarms	15 A	AFCI	14/2 + 14/3	panel → attic → JB-LT → each ceiling box; switch drops
18	LAUN-D	Gas dryer receptacle (laundry 2-gang, right)	15 A	Dual-function	14/2	panel → attic → straight to box
19	SPARE	Open — keep for future (e.g. a microwave or a 2-pole load needs both 19 + one more)				
20	ATTIC	Attic service receptacle + light + hatch switch	15 A	GFCI (210.8(E))	14/2	panel → attic → straight to box

Space 17 (LTG — room lights + smoke/CO alarms) is new; the 1 Nov E20 list had no whole-unit lighting circuit. Breakers are Eaton type BR: dual-function = BRN120DF / BRN115DF, AFCI = BRN120AF / BRN115AF, GFCI = BRN120GF — check the label in the panel door lists them.

Romex to buy

Cable	Unit (excl. kitchen)	Kitchen (KE Rev E)	Total on plan	+20%	Buy
12/2	603 ft	76 ft	679 ft	815 ft	**4 × 250 ft**
14/2	396 ft	162 ft	558 ft	670 ft	**3 × 250 ft**
14/3	65 ft	30 ft	95 ft	115 ft	**2 × 100 ft**

Each run = attic path between boxes (measured on the Magicplan plan, square to the walls) + the drop down the wall from a 121½ in ceiling + 10 in through joist and plate + 18 in tail at each end. Per-run list: 2026-11-02_U4_Electrical_Whole-Unit-Cable-Takeoff_R1.csv. Buy rolls, not cut lengths, so unopened extras go back. Home Depot Southwire SIMpull, checked 2 Nov 2026: 12/2 250 ft $182, 14/2 250 ft $125, 14/3 100 ft $117 — about $1,337 before tax for the rolls above.

Boxes and devices (unit, excluding kitchen)

Item	Qty	Where
Old-work outlet box, 1-gang (plaster walls)	20	L1–L6, H1, D1–D6, B1–B5, T1
Outlet box, 1-gang (open wall)	1	V1 (bath wall is open)
2-gang box, two circuits	1	Laundry: washer (20 A) + dryer (15 A), two receptacles, labeled
Switch boxes	8	Hall ×2 (3-way), living, dining, bedroom, laundry, bath (2-gang), attic hatch
Ceiling boxes	9	Hall ×2, living, dining, bedroom, laundry, bath light, 2 alarms
Bath exhaust fan with humidity sensor	1	Ducted outside; fed unswitched from JB-BL
Attic JB, 4-11/16 sq × 2-1/8 metal + blank cover	6	JB-DIN, JB-LIV (+ extension ring), JB-BED, JB-BR, JB-BL, JB-LT
Attic service receptacle (GFCI) + light	1 + 1	Next to the furnace; switch at the hatch
Receptacles 15/20 A tamper-resistant	≈ 21	TR everywhere (406.12); GFCI at V1 feeds T1
Smoke/CO combo + smoke alarm, hardwired, interconnected	1 + 1	Hall + bedroom; 14/3 between them

Code points that change the plan (2025 California Electrical Code + 2025 SF amendments)

• **Romex is allowed here.** SF 334.10(2) allows NM cable in multi-family wood-frame buildings up to 6 stories. Under 8 ft above the floor, SF treats wiring as open to damage (300.4(I)), so the cable stays inside walls and cabinets, never loose.

• **ENT (Smurf tube) in the walls is a problem in SF.** The 2025 SF amendment to 362.10 strikes most ENT uses and leaves concrete-encased uses. The 1 Nov rule (empty ENT first, no wire until inspection) is replaced here by NM-B in the walls, with the walls left open until rough inspection. Confirm with the inspector at rough.

• **Laundry:** one 20 A circuit with nothing else on it (210.11(C)(2)); GFCI + AFCI on both (210.8(A), 210.12(A)), so dual-function breakers. The laundry light goes on LTG, not on the laundry circuits.

• **Bathroom:** the outlets get their own 20 A circuit (210.11(C)(3)) and GFCI (210.8(A)(1)). Lights over the tub must be rated for damp or wet locations (410.10(D)).

• **Dining outlets** are a 20 A small-appliance circuit with no lights on it (210.52(B)). **Attic:** a GFCI service receptacle within 25 ft of the furnace, plus a light switched at the hatch (210.63, 210.70(A)(3)).

• **Permit:** SF only lets an owner pull an electrical permit for a single-family house. In this multi-unit building, a licensed electrical contractor pulls the permit and does the wiring (SF Bldg Code 89.120(E), 89.122(A)) — the same finding as the kitchen plan.

Still to settle

- Furnace and attic hatch: placed in the hall attic as an estimate. Move FURN, ATTIC REC and the hatch guard strips to the actual spots.
- Joist direction in the attic: run cable along the joist sides or bore through the joist centers. The 20% extra covers the longer way.
- D6 and L6 stay on hold until the pocket-door pier widths are known (no boxes in pocket cavities). B4/B5 and L5 are budgeted, not placed yet.
- Room ceiling lights in living, dining and bedroom: included so the whole unit runs off the new panel. Drop LTG branches for any room whose existing light wiring stays.
- Exact box marks come from the room drawings: Walls/Trim/Outlets R1, bedroom packet + Wall B Rev B, and kitchen KE Rev E.